GLaDOS

Geometric Lattice Deduction Over Streams

a sound, abstain-capable deduction organ — and the question of whether it survives being woven into a thinking LLM

ember + Claude (Opus 4.8) · draft 2026-06-25

Honesty banner. The sound-lattice-deduction + abstain + on-policy machinery is **not ours** — it is the Lattice Deduction Transformer (LDT, arXiv 2605.08605, May 2026). Our contribution is narrow and specific: **(a)** a Clifford geometric-product cell whose **wedge** is a pairwise-exclusion proposer, and **(d)** welding the deductor **inside** a pretrained LLM (OLMo-3), gradients through the solver, woven across layers. We write "empirically sound on this distribution," never "certified."

1 The one-sentence architecture

A Clifford transformer (the **cortex**) compiles a problem into a **constraint program** — initial candidate sets plus a pairwise-exclusion graph — and writes it into a recurrent **deductor** that runs sound powerset narrowing and returns a certified result or abstains. The cortex **reads it back** and continues. The spine:

$$a_{t+1} = a_t \cap \Pi_A(f_\theta(a_t, p, m))$$

where a_t is a point in a per-position powerset lattice, Π_A is monotone narrowing (the meet keeps the result $\subseteq a_t$), and f_θ is the geometric-product cell. Bilinear **proposes**; the lattice meet is the sole **authority** on elimination.

1.1 Why Clifford is not arbitrary

The wedge $u \wedge v$ is antisymmetric ($u \wedge u = 0$): it discards self-magnitude and keeps only cross-channel structure — exactly "these two candidates cannot coexist," the primitive a constraint graph is built from. A Clifford cortex computes, as its native representation op, the same algebra the deductor consumes as constraints. The host's wedge features become the deductor's exclusion edges. Deep coupling, not vocabulary.

2 What is ours vs what is borrowed

ingredient	**source**	**status**
(b) lattice / abstract-interpretation narrowing / abstain	LDT 2605.08605	**not novel**
(c) on-policy alpha-target on own deduced states	LDT 2605.08605	**not novel**
(a) Clifford wedge = pairwise-exclusion deduction op	ours; cf. CliffordNet 2601.06793, Budinich 1704.02942	**partially novel, narrow**
(d) sound deductor woven **inside** a pretrained LLM, grad through solver	ours	**novel as a combi-**

ingredient	source status
	na-tion

The honest reduction: LDT's leap to 100% on Sudoku-Extreme came from the **symbolic lattice**, not neural cleverness (cf. TRM, 7M params, 87.4%, arXiv 2510.04871). So our entire bet is two questions — **does the wedge add anything on top of the lattice?** and **does it survive the LLM graft?**

3 The decisive experiment (Layer 0)

Iso-parameter core-swap on Sudoku-Extreme (≈800K params, identical solve loop, only the channel mixer varies):

- **ldt** — attention + SwiGLU FFN. This is **also the control the HRM critics demand** (a plain looped transformer at equal refinement count; ARC-Prize showed HRM's gains were the refinement loop + augmentation + puzzle-id memorization, not architecture).
- **glados** — attention + geometric product (dot + wedge), no FFN.
- **nowedge** — ablation: dot-only, the antisymmetric branch removed.

3.1 Metric that decides it

False-elimination rate — the fraction of (non-given, true-still-alive) cells where the meet just killed the true candidate. A sound deductor's value is ≈ 0. Plus solve rate, **wrong-return rate** (soundness violations at output), and p90 forward-passes per solved puzzle. Decision rule: glados **has legs** iff it matches/beats ldt's solve rate at equal params **and** lowers false-elimination (or cuts p90 forwards ≥ 25%). Otherwise the wedge is decoration and Clifford is just an efficiency primitive.

Results — Sudoku-Extreme, iso-param, 2000 steps. *pending the running 3-way*

arm	params	false-elim	solve	wrong-ret
ldt	—	—	—	—
glados	—	—	—	—
nowedge	—	—	—	—

4 Beyond the organ

1. **Domain-general gauntlet** (built): one organ, many lattices — graph-coloring, 3-SAT, maze through one interface; hold out a domain, test **zero-shot soundness**. The honest form of "generalizes to nonsense."
2. **The co-processor** (Layer 1): zero-init gated graft into OLMo-3-Base-7B (Flamingo-style tanh gate → step-0 hybrid is OLMo exactly); baseline to beat = OLMo-3-Think-7B. Kill-risk: does the cortex **learn to emit valid constraint programs**, or route around the organ?
3. **Streams & lifting**: semigroup binary-lifting $F_2 \approx \Pi(F_1 \circ F_1) \to O(\log T)$ deduction; the lifting error is itself a diagnostic of how shortcuttable the task's deduction semigroup is.
4. **Online edge-of-stability**: cheap running proxies (participation ratio, finite-time Lyapunov, top-k Hessian) the whole way along. Our hypothesis-under-test: **attractor dimension predicts cross-domain soundness better than val-loss**.

5 Open honesty / risks

- "Sound" is **empirical**, on-distribution. We have no proof and no in-loop verifier yet. Even LDT breaks soundness on Maze-Hard.

- (a) is at risk of "that's just the Pauli analogy renamed" — read Budinich + CliffordNet before claiming.
- Must reproduce the plain-looped-transformer control (our ldt arm) and cite LDT/TRM head-on.

References: see `notes/prior_art.md` *for the full lineage and URLs (LDT 2605.08605, TRM 2510.04871, ARC-Prize HRM analysis, CliffordNet 2601.06793, SATNet 1905.12149, …).*